

Supplemental Methods: STATE

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We benchmarked the State Embedding (“SE”) stage of STATE[1], a self-supervised cell-embedding transformer trained with a tabular optimal-transport-style loss (Wasserstein + energy kernel) over masked gene-expression “sentences”. Each cell is tokenized into a sequence of (gene, count) pairs, gene tokens are mapped to fixed protein-language-model embeddings, and the sequence is processed by a transformer encoder; a binary decoder simultaneously predicts gene presence to encourage sparse reconstruction.

Following the published recipe, gene tokens are replaced with pretrained protein embeddings from TranscriptFormer[2], a cross-species single-cell foundation model. We use the 2,560-dimensional gene embeddings released by TranscriptFormer for both *Mus musculus* and *Homo sapiens*, merged into a single multi-species table and matched to each dataset’s gene names via Ensembl identifier resolution.

To make STATE comparable to the Geneformer baseline used elsewhere in this study, we override the published SE defaults to a smaller architecture (Table 1): 256 hidden, 4 attention heads, 3 transformer layers, 512 feed-forward, 256-dim cell-embedding output, dropout 0.02. Because gene tokens are mapped to a fixed embedding table, the trainable parameter count of SE under our configuration is dominated by the encoder, transformer, and decoder ($\sim 3.5\text{--}4\text{M}$ trainable parameters across datasets), rather than by a learned gene-vocabulary embedding.

Unlike Geneformer and SCVI, STATE training duration is governed by a fixed *step* budget chosen so that the total amount of optimization remains roughly constant across the sweep, with step-based early stopping (patience = 5 validation windows) acting as an upper bound. After training, the best checkpoint is selected by minimum validation loss, and cell embeddings are extracted from this checkpoint for downstream mutual-information estimation.

Figure 1 shows representative training and validation loss curves for SE trained on PBMC at the largest dataset size (46,415 cells) and full quality. Both losses decrease monotonically and the validation curve closely tracks training, indicating stable convergence without overfitting.

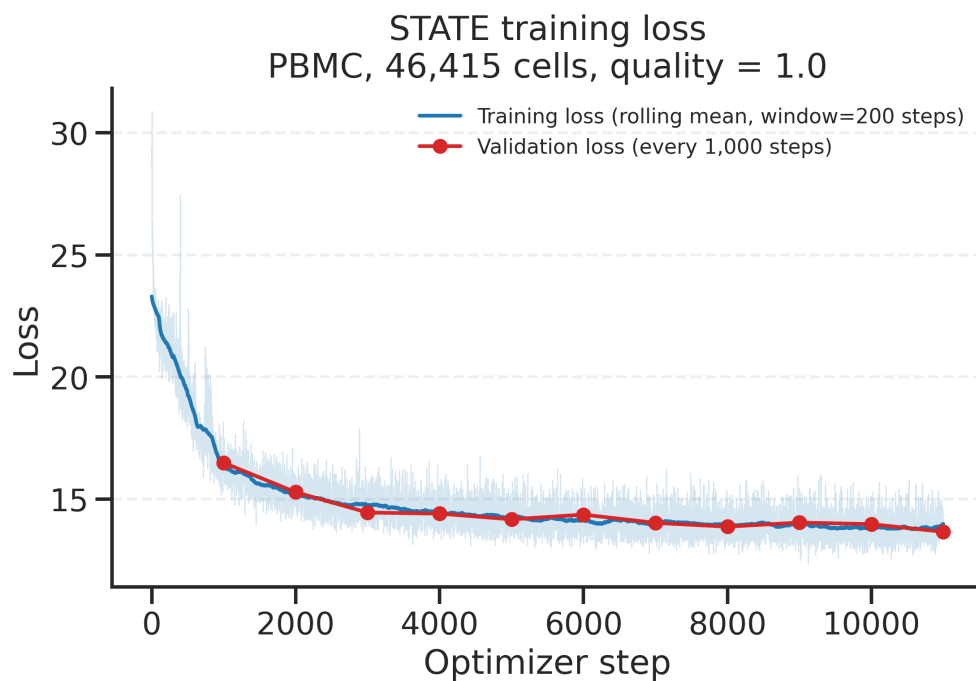


Figure 1: Training and validation loss curves for STATE SE trained on PBMC (46,415 cells, quality = 1.0), shown on linear axes. Validation is computed every 1,000 optimizer steps. The raw per-step training loss is plotted faintly in the background, and the bold blue curve overlays a centered rolling mean (window = 200 steps) to suppress minibatch noise; no other smoothing is applied. The validation curve is shown unsmoothed (one point per evaluation).

Table 1: Hyperparameters for STATE (SE) as used in this study.

Parameter	Value
<i>Model architecture</i>	
Number of transformer layers	3
Hidden dimension	256
Attention heads	4
Feed-forward dimension	512
Cell-embedding (output) dimension	256
Maximum sequence length	2,048 tokens
Dropout	0.02
Gene token embedding	TranscriptFormer protein embeddings (2,560-dim, mouse + human)
Token-level objective	20% variable masking
Reconstruction head	Binary gene-presence decoder + count head
Loss	Tabular (Wasserstein + energy kernel)
<i>Optimization</i>	
Optimizer	AdamW
Maximum learning rate	1×10^{-4}
Weight decay	0.01
Maximum gradient norm	0.8
Global batch size	64
LR schedule	OneCycleLR (warmup 0.33, end 1.0)
<i>Training and checkpointing</i>	
Validation interval	1,000 optimizer steps
Validation batches per check	50
Best-checkpoint selection	Minimum validation loss
Early-stopping patience	5 validation windows ($\geq$ 1-epoch warm-up)

References

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